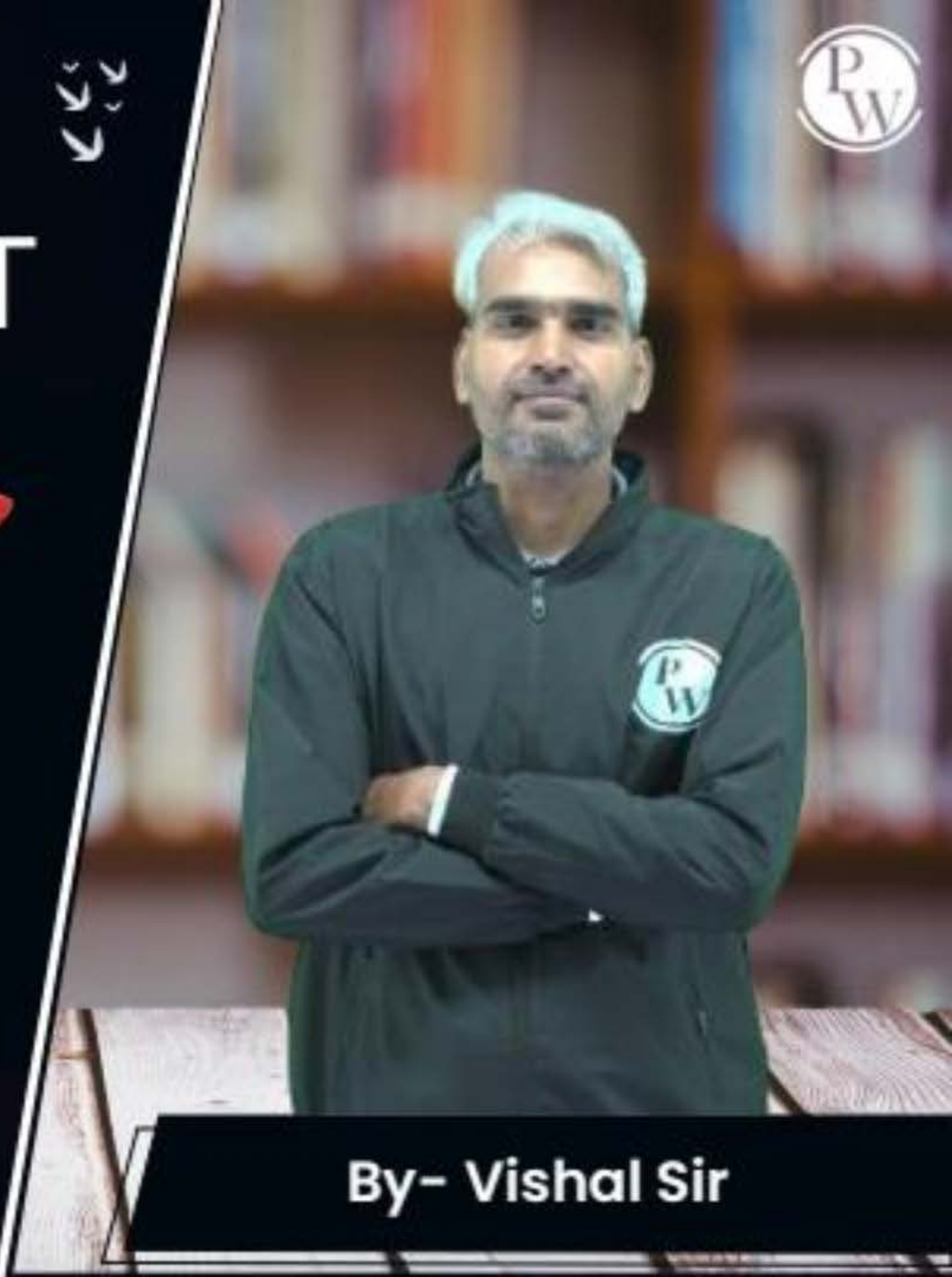
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 02



By- Vishal Sir

Recap of Previous Lecture



Topic Introduction to Graph Theory

Topic Representation of graph

$G = (V, E)$

$e_1 = \{v_i, v_j\}$ or $e_1(v_i, v_j)$

Topic Simple graph, multi-graph and pseudo graph

No loop + No parallel edges

Topic Maximum number of edges in a simple graph with n vertices $^nC_2$

Topic Number of simple graphs possible with n labeled vertices $2^{^nC_2}$

Topic Degree of a vertex and terminologies associated with it

Slide

Topics to be Covered



Topic Some important terminologies

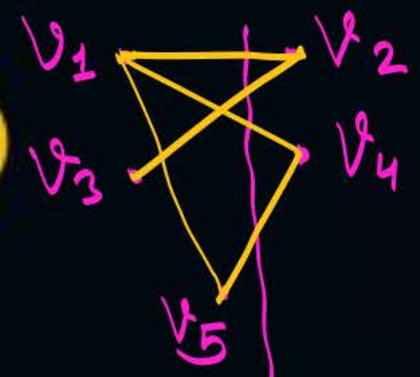
Topic Different types of graphs

Slide

Topic : Graph

* Undirected graph:-





$e_1 = \{v_1, v_2\}$
$e_2 = \{v_1, v_4\}$
$e_3 = \{v_1, v_5\}$
$e_4 = \{v_2, v_3\}$
$e_5 = \{v_4, v_5\}$

$G_1 = (V_1, E_1)$

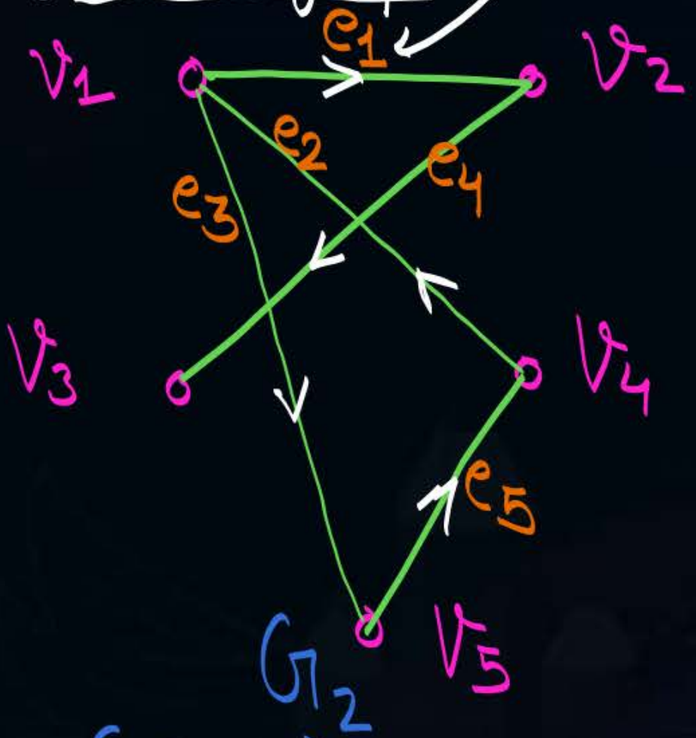
$V_1 = \{v_1, v_2, v_3, v_4, v_5\}$

$E_1 = \{e_1, e_2, e_3, e_4, e_5\}$

$E_2 = \{\{v_1, v_2\}, \{v_1, v_4\}, \{v_1, v_5\}, \{v_2, v_3\}, \{v_4, v_5\}\}$

Directed graph:

edge is from v_1 to v_2



$\neq (v_2, v_1)$

$e_1 = (v_1, v_2)$
$e_2 = (v_4, v_1)$
$e_3 = (v_1, v_5)$
$e_4 = (v_2, v_3)$
$e_5 = (v_5, v_4)$

$G_2 = (V_2, E_2)$

$V_2 = \{v_1, v_2, v_3, v_4, v_5\}$

$E_2 = \{e_1, e_2, e_3, e_4, e_5\}$



Topic : Graph

A Graph is defined as an ordered pair of two sets,

order in which elements appear is important

Set of vertices & Set of edges

i.e., $G = (V, E)$

Name of graph

Set of vertices

Set of edges

{edges are used to define that how exactly vertices of the graph are connected with each other}

$|V|$ = Numer of vertices in set V ($|V|$ = Order of graph)

$|E|$ = Number of edges in set E ($|E|$ = Size of graph)



Topic : Representation of edge

- In a non-directed graph an edge is represented by a set of two vertices

  e.g., {vi, vj} which represents an edge between vertices vi and vj

  $= \{v_j, v_i\}$

  v_i & v_j are called end-vertices associated with that edge

- In a directed graph an edge is represented by an order pair of two vertices

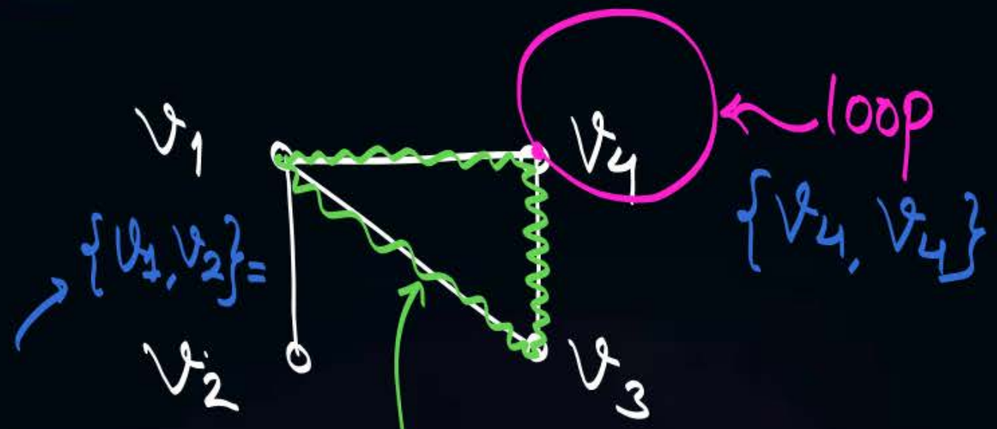
  e.g., (vi, vj) which represents an edge from vertex vi to vertex vj

  $\neq (v_j, v_i)$

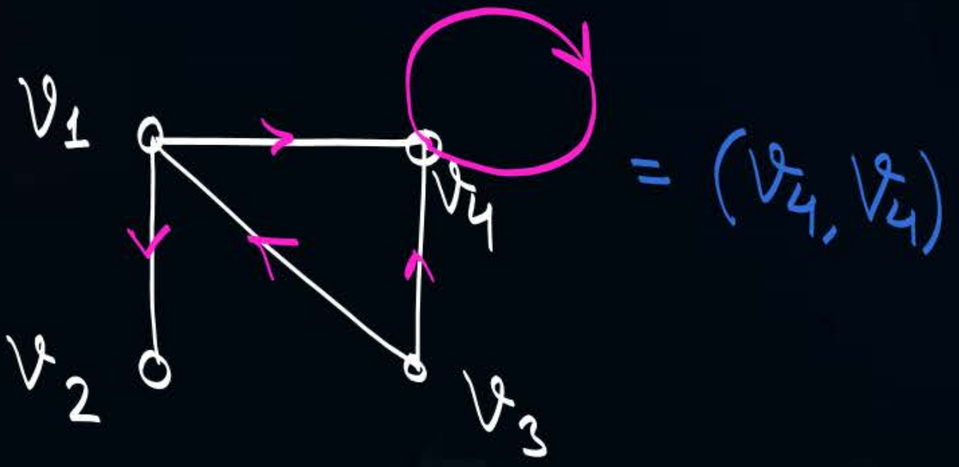


Topic : Loop

Self-loop :-



- it is not a loop it is called Cycle in the graph



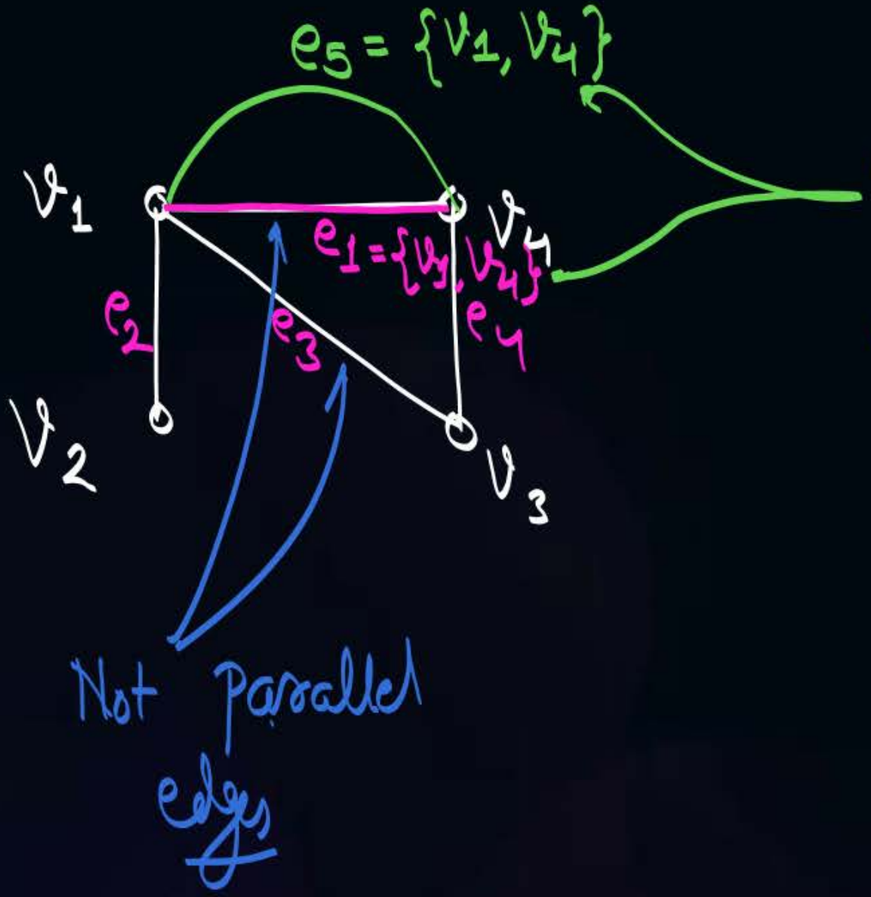


Topic : Loop

both

An edge in which end vertices associated with the edge are same is called as loop or self-loop.

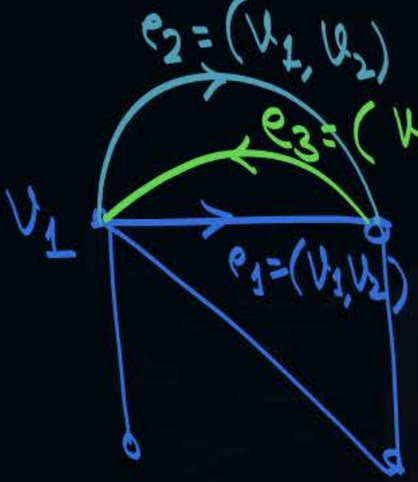
Topic : Parallel Edges



Not parallel edges

Parallel edges exist

∴ Graph is a multi-graph



it is not parallel to e_1 and e_2



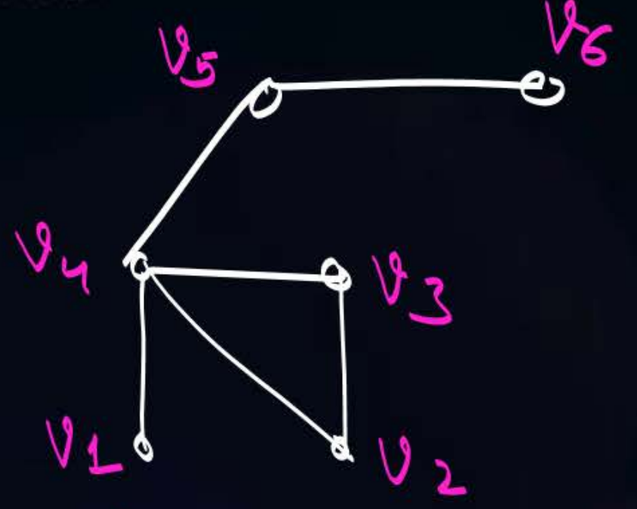
Topic : Parallel Edges

If two or more edge are associated with same end vertices, then edges are called parallel edges and resulting graph is called multi-graph.



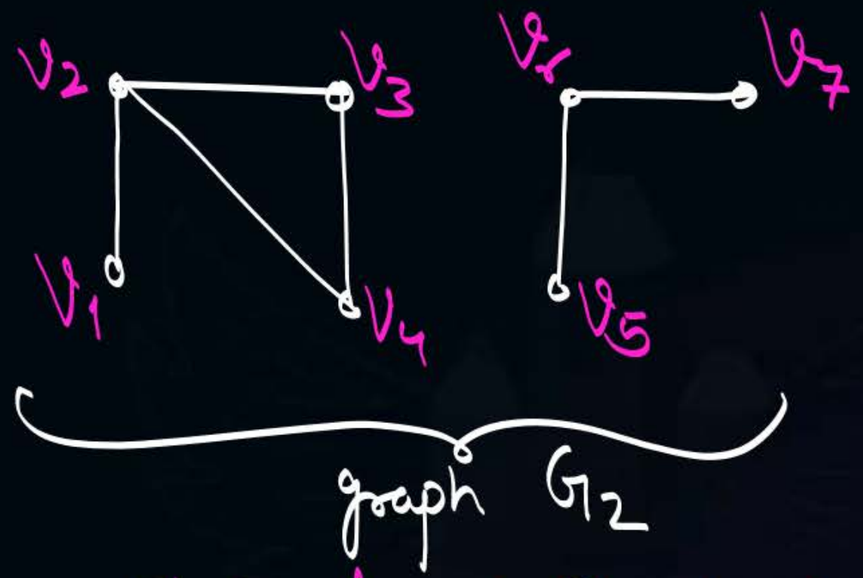
Topic : Simple graph

A graph with no self-loop and no parallel edge is called simple graph,



G_1

It is a simple graph as well as Connected graph



graph G_2

It is also a simple graph but it is a dis-connected graph



Topic : Classification of graph

	Loop	Parallel edge
Simple Graph	Not allowed	Not allowed
Multi-graph	Not allowed	Allowed
Pseudo graph	Allowed	Allowed



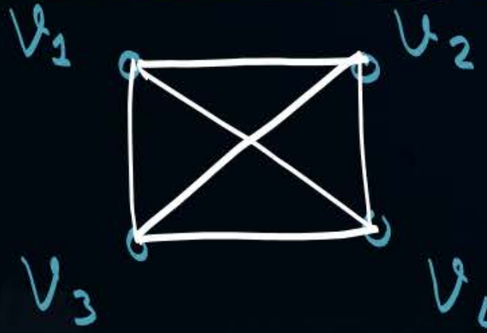
Topic : Simple Graph

Maximum number of edges possible in "Simple Graph" With n vertices (labeled)

* In a simple graph every edge will be between two distinct vertices of the graph

* ∴ Maximum No. of edges Possible in a simple graph with n-vertices = No. of ways in which two distinct vertices can be choosen out of n distinct vertices = $^nC_2$

Maximum no. of edges Possible in a simple graph with n-vertices = $^nC_2 = \frac{n(n-1)}{2} = \frac{n^2-n}{2}$



$\{v_1, v_2\} = \{v_2, v_1\}$
$\{v_1, v_3\} = \{v_3, v_1\}$
$\{v_1, v_4\} = \{v_4, v_1\}$
$\{v_2, v_3\} = \{v_3, v_2\}$
$\{v_2, v_4\} = \{v_4, v_2\}$
$\{v_3, v_4\} = \{v_4, v_3\}$



Topic : Simple Graph

labeled

Maximum number of edges possible in "Simple Graph" With n-vertices

In a simple graph with n-vertices
number of edges will always be $\leq \frac{n(n-1)}{2}$

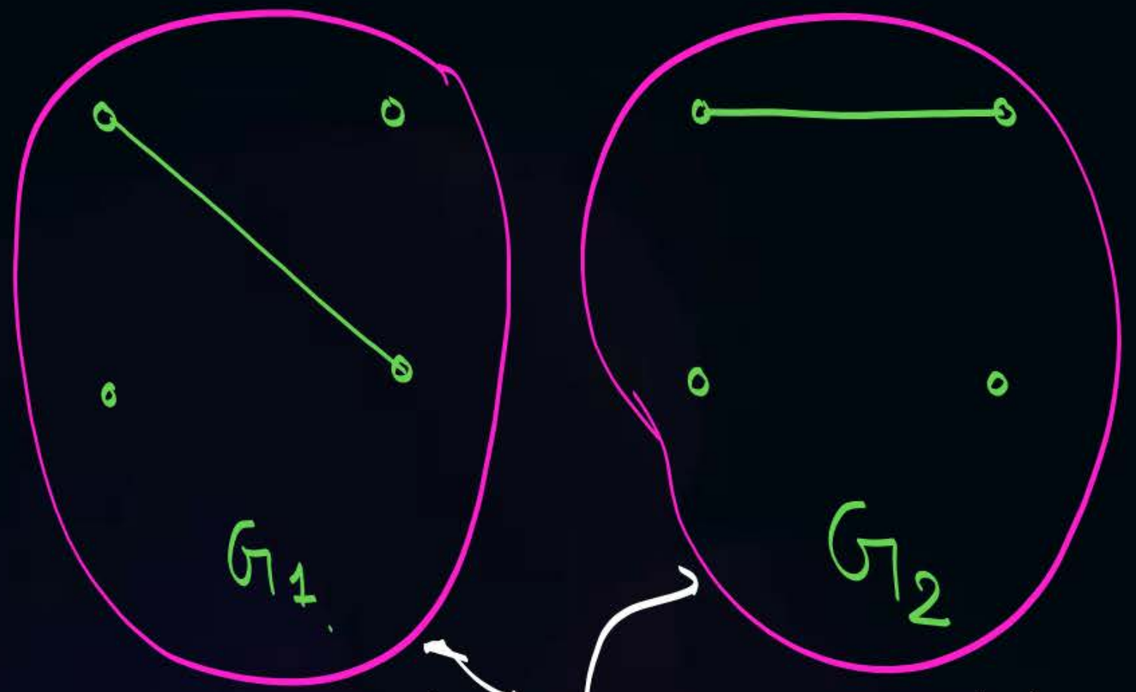


$\{v_1, v_2\} = \{v_2, v_1\}$
$\{v_1, v_3\} = \{v_3, v_1\}$
$\{v_1, v_4\} = \{v_4, v_1\}$
$\{v_2, v_3\} = \{v_3, v_2\}$
$\{v_2, v_4\} = \{v_4, v_2\}$
$\{v_3, v_4\} = \{v_4, v_3\}$



Topic : Simple Graph

Number of "Simple Graphs" possible With 'n' labeled vertices



If vertices are not labeled, then both the graphs are same



If vertices are labeled then graphs are different



Topic : Simple Graph

Number of "Simple Graphs" possible With 'n' labeled vertices

In a simple graph with n-vertices there will be exactly n-vertices and at most $^nC_2$ edges

i.e A simple graph with n-vertices may have

n-vertices & 0-edges (OR) n-vertices & 1-edge (OR) n-vertices & 2-edges (OR) . - - - (OR) n-vertices & all $^nC_2$ edges



Topic : Simple Graph

Number of "Simple Graphs" possible With 'n' labeled vertices

If vertices are labeled, then all $^nC_2$ edges are different from each other.

Number of simple graph possible with n-labeled vertices

= Choose all n-vertices and { Choose '0' out of $^nC_2$ edges (or) Choose '1' out of $^nC_2$ edges (or) Choose '2' out of $^nC_2$ edges - - - - (or) Choose all $^nC_2$ edges out of $^nC_2$ edges }

$= {}^nC_n * \{ {}^{nC_2}C_0 + {}^{nC_2}C_1 + {}^{nC_2}C_2 + \cdots + {}^{nC_2}C_k + \cdots + {}^{nC_2}C_{nC_2} \} = 1 \cdot 2^{nC_2} = 2^{\frac{n(n-1)}{2}}$



Topic : Simple Graph

Number of "Simple Graphs" possible With 'n' labeled vertices

No. of simple graphs possible with n-labeled vertices

$= {}^{n_{C_2}}C_0 + {}^{n_{C_2}}C_1 + \cdots + {}^{n_{C_2}}C_k + \cdots + {}^{n_{C_2}}C_{n_{C_2}} = 2^{n_{C_2}} = 2^{\frac{n(n-1)}{2}}$

No. of simple graphs with n-labeled vertices and at most 'K' edges

No. of simple graphs Possible with n-labeled vertices & Exactly K-edges

No. of simple graphs Possible with n-labeled vertices & at least 'K' edges

Q: How many simple graphs are possible with '5' labeled vertices

Soln: Maximum edges possible $= {}^5C_2 = 10$

∴ No. of simple graphs possible $= 2^{10} = \underline{1024}$

Q. How many simple graphs are possible with 4 labeled vertices & at most '2' edges.

Soluⁿ:- Maximum No. of edges possible in a simple graph with 4 vertices $= {}^4C_2 = 6$

&

No. of simple graphs possible with all 4-vertices & at most '2' edges $= {}^4C_4 * \{ {}^{6}C_0 + {}^{6}C_1 + {}^{6}C_2 \}$

$= 1 * \{ {}^{6}C_0 + {}^{6}C_1 + {}^{6}C_2 \}$

$= 1 + 6 + 15 = \boxed{22}$ Ans

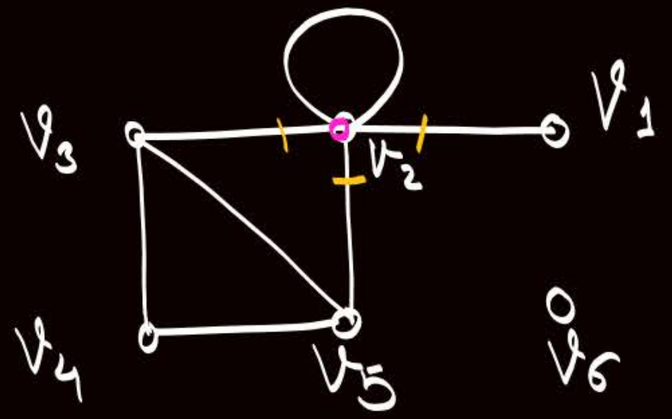


Topic : Degree of a vertex in an Undirected Graph

In an undirected graph, degree of a vertex 'v' is denoted by "deg(v)" and it is defined as

$deg(v)$ = Number of edges incident with vertex 'v'

Eg:



$\delta(G) = 0$

$\Delta(G) = 5$

$\deg(v_1) = 1$ { A vertex with degree = 1 is called Pendant vertex }

$\deg(v_3) = 3$

$\deg(v_4) = 2$

$\deg(v_5) = 3$

$\deg(v_6) = 0$ { A vertex with degree = 0 is called an isolated vertex }

Note: In an undirected graph a loop at a vertex is counted as two edges, to obtain the degree of that vertex

$\therefore \deg(v_2) = 3 + 2 = 5$

3 — w.r.t. simple edges

2 — w.r.t. loop



Topic : Degree of a vertex in directed Graph

In a directed graph two types of degrees are defined for each vertex of the graph

(1) In-degree

(2) out-degree



Topic : Degree of a vertex in directed Graph

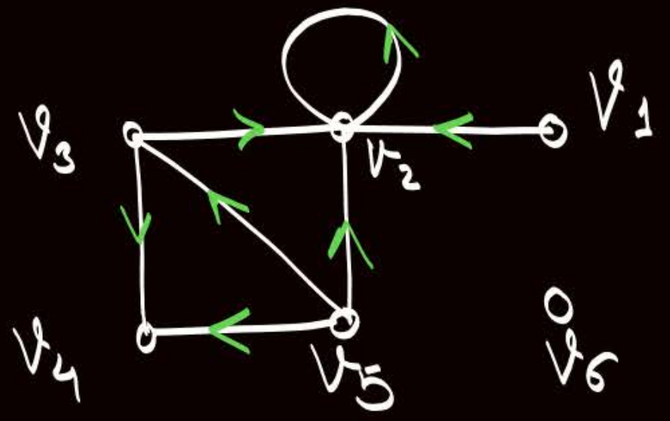
① In-degree :- In a directed graph in-degree of vertex 'v' is denoted by $deg^+(v)$, and it is defined as

$deg^+(v)$ = No. of edges incoming to vertex v.

② Out-degree :- In a directed graph out-degree of vertex 'v' is denoted by $deg^-(v)$, and it is defined as

$deg^-(v)$ = No. of edges out-going from vertex 'v'.

Eg:



Note: In a directed graph a loop at a vertex is counted as one edge to obtain the indegree of that vertex & counted as one edge to obtain out-degree of that vertex

$\deg^+(v_1) = 0$, $\deg^-(v_1) = 1$, $\Sigma = 1 = \deg(v_1)$

$\deg^+(v_3) = 1$, $\deg^-(v_3) = 2$, $\Sigma = 3 = \deg(v_3)$

$\deg^+(v_4) = 2$, $\deg^-(v_4) = 0$, $\Sigma = 2 = \deg(v_4)$

$\deg^+(v_5) = 0$, $\deg^-(v_5) = 3$, $\Sigma = 3 = \deg(v_5)$

$\deg^+(v_6) = 0$, $\deg^-(v_6) = 0$, $\Sigma = 0 = \deg(v_6)$

$\deg^+(v_2) = 3 + 1 = 4$ (w.r.t. loop)

$\deg^-(v_2) = 0 + 1 = 1$ (w.r.t. loop)

$\Sigma = 5 = \deg(v_2)$

- Summation of in-degree and out-degree of a vertex in directed graph is defined as overall degree of that vertex

i.e. $$\deg(v) = \deg^+(v) + \deg^-(v)$$



Topic : Null Graph

A graph with no edge in it is called a null graph.

{ In a null graph vertices may exist but edges can not exist }

In a graph $G = (V, E)$
if $|E| = 0$ then
graph 'G' is a Null graph

If $deg(v) = 0$, $\forall v \in G$
then 'G' is a Null graph

Vertices may exist without edges, but edges can not exist without vertices

Note:- A graph with no vertex and no edge is called an "Empty graph"



Topic : Isolated Vertex

A vertex 'V' with degree = 0 is called an isolated vertex



Topic : $\delta(G)$ and $\Delta(G)$

In a graph G,

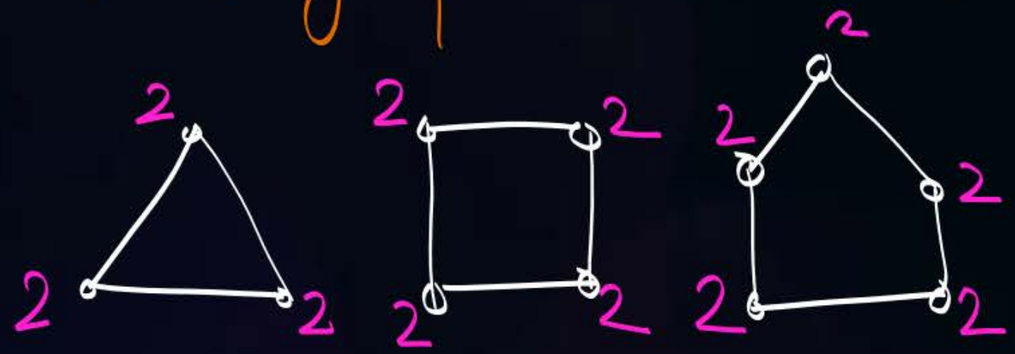
$\delta(G)$ = Minimum of the degrees of all the vertices in graph G is denoted by $\delta(G)$

$\Delta(G)$ = Maximum of the degrees of all the vertices in graph G is denoted by $\Delta(G)$

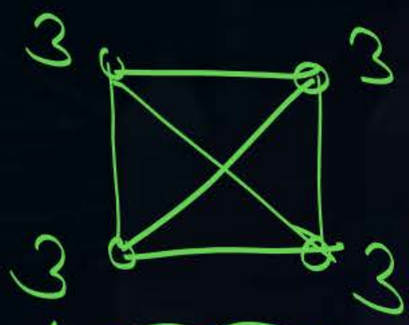


Topic : Regular Graphs

- In a graph if degree of each vertex is same then graph is called a regular graph.
- In a graph if degree of each vertex is exactly 'k' then graph is called 'k-regular' graph



Every Polygon is a '2-regular' graph

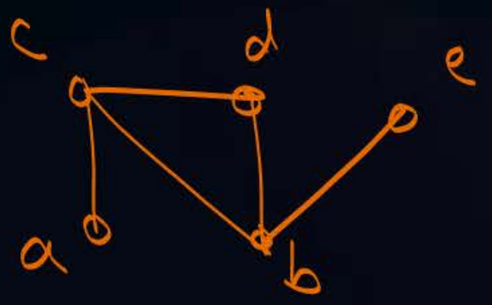


it is a '3-regular' graph



Topic : Adjacent vertices and Adjacent edges

Two vertices are said to be adjacent if there exist a direct edge between them

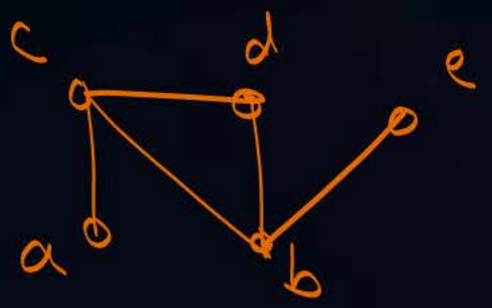


b & d are adjacent
b & c are adjacent
a & d are not adjacent
a & b are not adjacent



Topic : Adjacent vertices and Adjacent edges

Two edges are said to be adjacent if they share a common vertex



$\{a, c\}$ & $\{b, c\}$ are adjacent edges

$\{b, c\}$ & $\{b, e\}$ are adjacent edges

$\{a, c\}$ & $\{b, d\}$ are not adjacent



Topic : Types of graphs

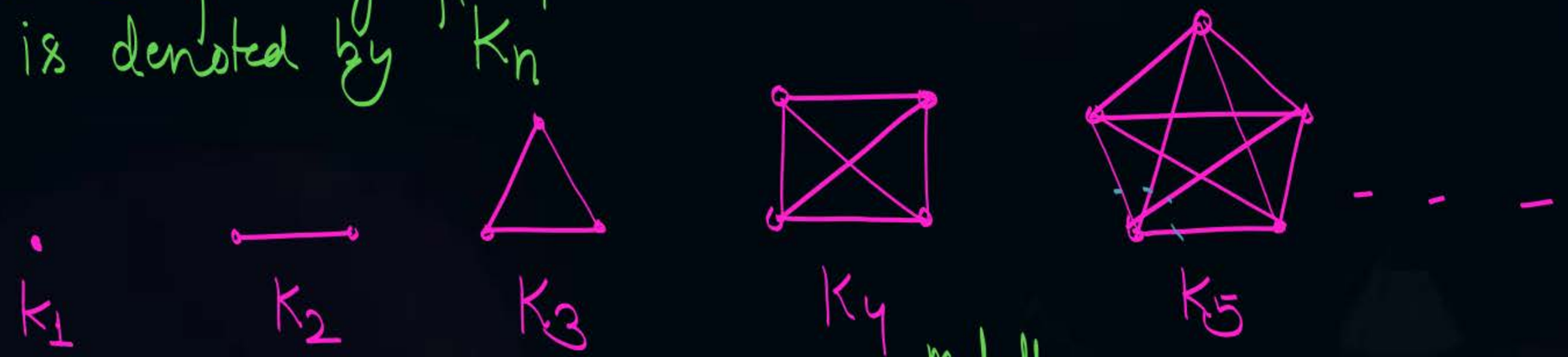
1. Complete Graph
2. Cycle Graph
3. Wheel Graph
4. Connected Graph
5. Cyclic Graph
6. Acyclic Graph
7. Tree
8. Bipartite Graph
9. Complete Bipartite Graph



Topic : Complete Graph

A simple graph with all possible edges is called a Complete graph

* A Complete graph with n-vertices is denoted by 'K_n'



* A simple graph in which all vertices are mutually adjacent, is called a Complete graph

* A Complete graph K_n is a simple graph with n-vertices & all possible edges

$$\therefore \boxed{|E(K_n)| = {}^{n}C_2 = \frac{n(n-1)}{2}}$$

* In a Complete graph K_n, degree of each vertex is '$n-1$', $\therefore$ Complete graph K_n is a '$(n-1)$-regular' graph



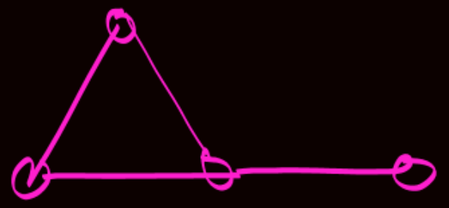
Topic : Cycle Graph



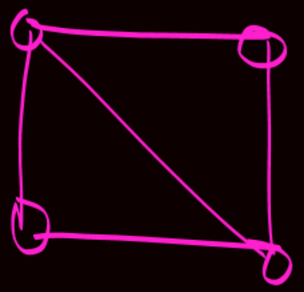
* A simple graph with n-vertices ($n \geq 3$), where all 'n' vertices form a cycle of length = n is called a cycle graph.
* A cycle graph with n-vertices is denoted by C_n.



* length of a cycle is equal to the number of edges in that cycle
* $|E(C_n)| = n$
* Every cycle graph is a '2-regular' graph



Not a
cycle graph



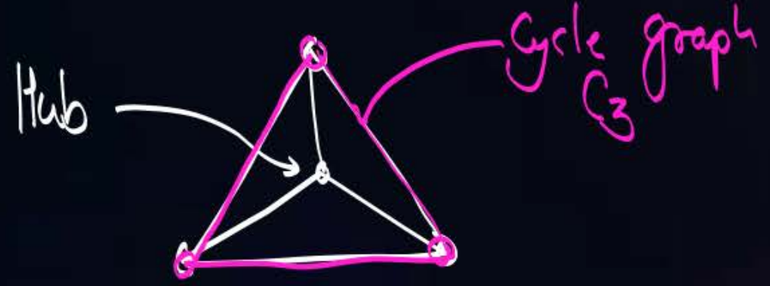
Not a
cycle graph



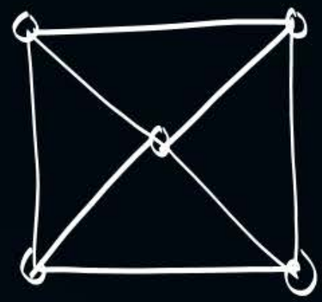
Topic : Wheel Graph

* A wheel graph with n-vertices is denoted by W_n. *

A wheel graph with 'n' vertices can obtained by connecting all the vertices of a cycle graph C_{n-1}, with a new vertex called hub vertex



$W_4 = C_3 + 1\text{-hub}$



$W_5 = C_4 + 1\text{-hub}$

Find the no. of edges in wheel graph W_n = ?

$$|E(W_n)| = (n-1) + (n-1)$$

$(n-1)$: w.r.t. C_{n-1}

$(n-1)$: to Connect hub vertex with $(n-1)$ vertices of C_{n-1}

$$\boxed{|E(W_n)| = 2(n-1) = 2n-2}$$

- In a Wheel graph W_n
  - (i) degree of hub-vertex = $(n-1)$
  - (ii) degree of vertices at cycle = 3

* A Wheel graph W_n can be a regular graph if and only if $(n-1) = 3$

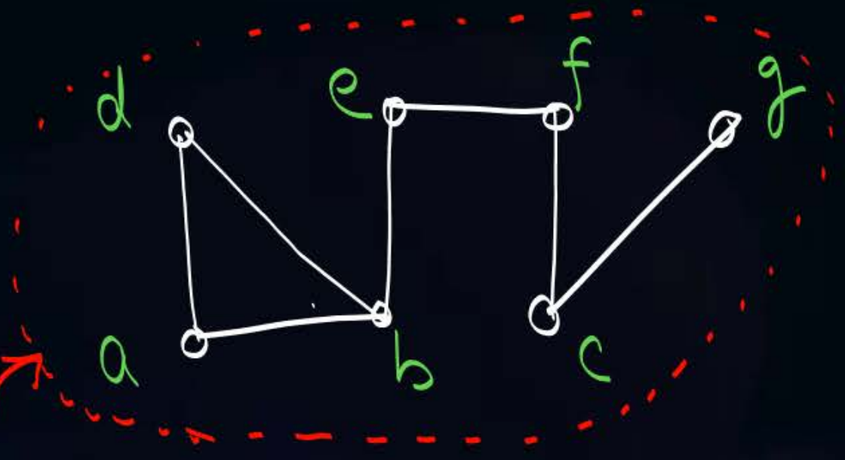
i.e. $\boxed{n = 4}$ ✓

The only wheel graph which is also a regular graph is W_4.



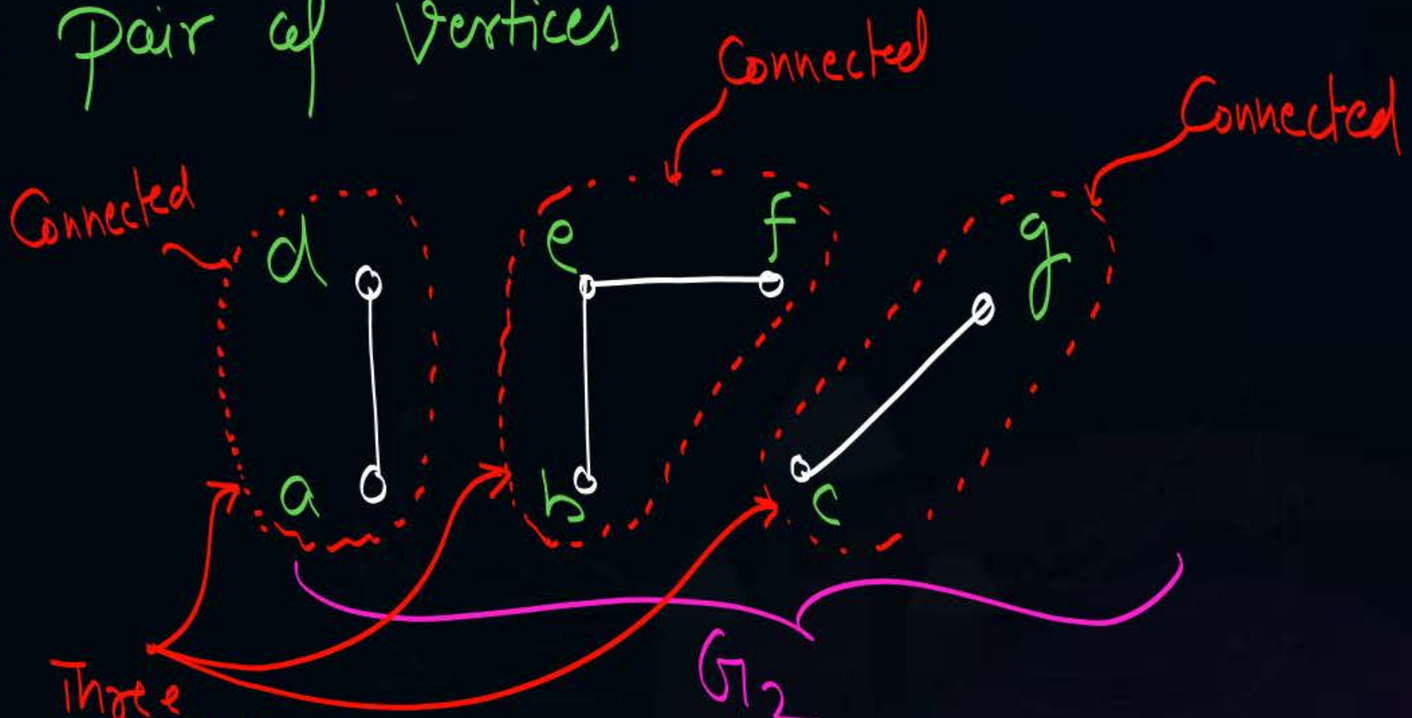
Topic : Connected Graph

A graph is said to be connected if there exist a path between every pair of vertices



A single Connected Component

it is a Connected graph



Three Connected Components in this graph

G_2

G_2 is a disconnected graph

- In a Connected graph there will be only one Connected Component.
- In a disconnected graph there exist two or more Connected Components
- K_n, C_n, & W_n are always Connected graphs
- Connected graph may or may not be Simple



Connected but not simple



Topic : Cyclic Graph

- A graph which contains at least one cycle in it is called a cyclic graph

eg



Contains a cycle

∴ Cyclic graph as well as Connected

eg



G

It is disconnected graph but a cyclic graph

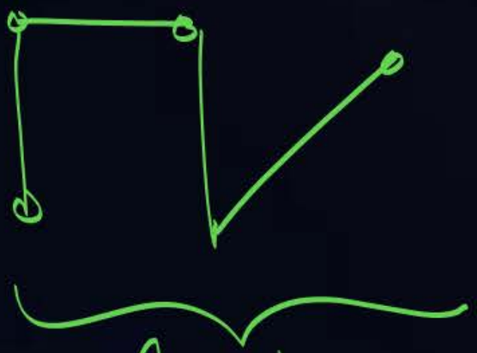
* Every cycle graph is a cyclic graph, but Every cyclic graph need not be a cycle graph.

* A cyclic graph may or may not be connected



Topic : Acyclic Graph

- A graph with no cycle in it is called an acyclic graph



Acyclic
+
Connected

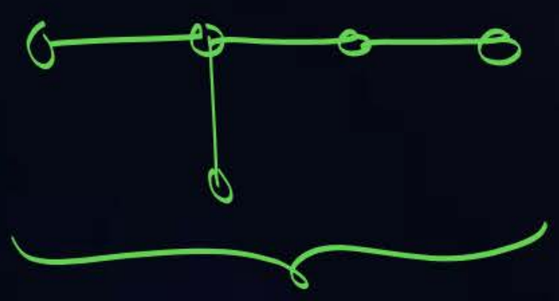


Acyclic
+
disconnected



Topic : Tree

- A tree is an acyclic connected graph
- A tree with 'n' vertices will have exactly (n-1) edges.



Tree

A collection of tree is called forest

Note:- ① A tree is a connected graph with minimum number of edges

② A graph with n vertices and less than $(n-1)$ edges is always a disconnected graph

③ A graph with n vertices and no. of edges $\geq (n-1)$ may or may not be connected

④ A simple graph with n vertices and more than $(n-1)$ edges ($> (n-1)$) is always cyclic

⑤ A graph with n-vertices and no. of edges $\leq (n-1)$ then graph may or may not be cyclic

H.W: A forest with n-vertices & k-connected components will have how many edges ?



Topic : Bi-partite Graph

A bipartite graph (or bigraph) is a graph whose vertices can be partitioned into two sets V_1 and V_2 such that every edge of the graph connects a vertex in set V_1 to a vertex in set V_2.

$V_1 \cap V_2 = \emptyset$

$V_1 \cup V_2 = V$

No edge should connect a vertex of set V_1 to another vertex of V_1 and No edge should connect a vertex of set V_2 with another vertex of set V_2

A bipartite graph can be represented using the notation $G = (V_1, V_2, E)$ where,

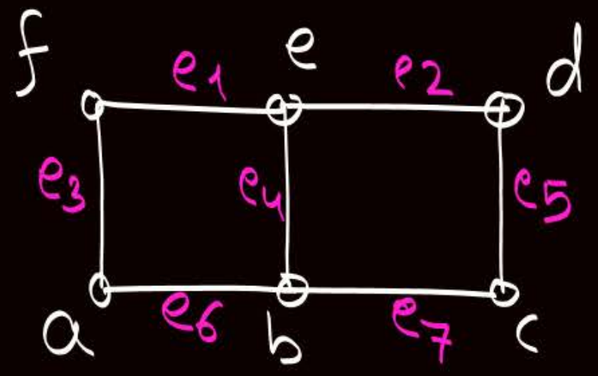
ordered triple

i.e No two vertices of set V_1 should be adjacent to each other & No two vertices of set V_2 should be adjacent to each other

V_1 is the first set of the partition

V_2 is the second set of the partition and

E is the set of edges of graph G



G

$V = \{a, b, c, d, e, f\}$

$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$

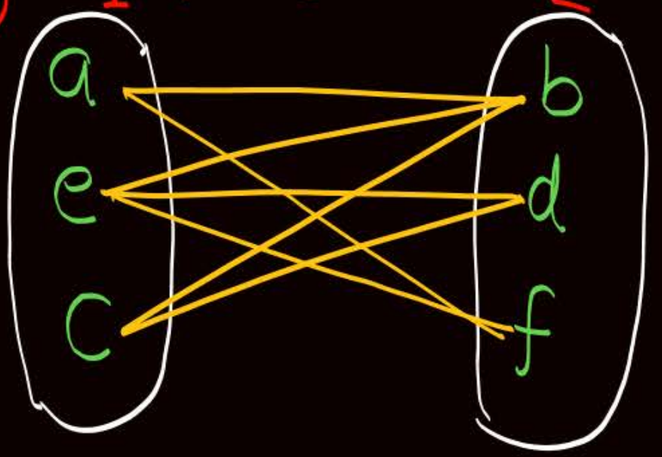
Partition of V $= \{\{a, c, e\}, \{b, d, f\}\}$

$V_1 = \{a, e, c\}$ $V_2 = \{b, d, f\}$

$V_1 \cap V_2 = \emptyset$

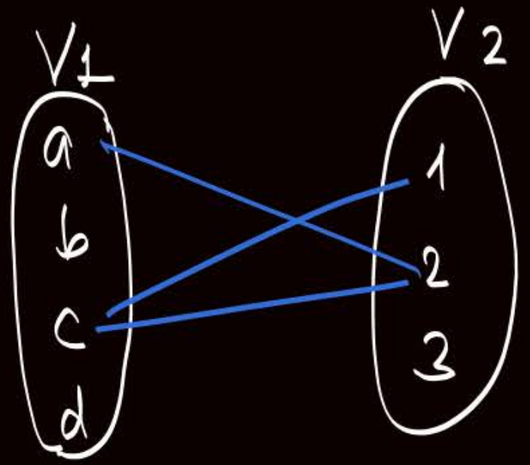
& $V_1 \cup V_2 = V$

∴ it is a Partition of V

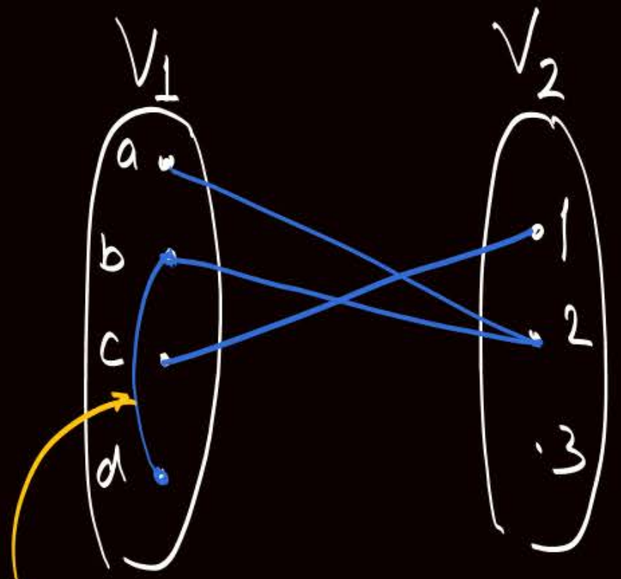


We can partition the set of vertices of graph G into two subsets V_1 & V_2 such that every edge of graph G connects a vertex of set V_1 with a vertex of set V_2

∴ G is a bipartite graph



it is a bipartite graph



two vertices of the same subset of partition are adjacent to each other

∴ it is not a bipartite graph



2 mins Summary

Topic Some important terminologies

Topic Different types of graphs

THANK - YOU